



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Odd Semester)

| VISION                                                                                       | MISSION                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To emerge as a learning hub and center of excellence in the domain of Information Technology | [M1] To impart quality technical education through effective teaching learning process.<br>[M2] To provide a platform to address societal issues as well as challenges faced by IT industries.<br>[M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT.<br>[M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning. |

### Class Test (CT)-2

Semester: V

Programme: B. Tech Information Technology

Course: Data Warehousing and Mining

Time: 1 Hr 60 min

Max Marks: 30 Marks

#### Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

| BIT33505 | Course Outcome                                                                                                                     |
|----------|------------------------------------------------------------------------------------------------------------------------------------|
| CO1      | To classify the fundamentals, need, structure, and cost of data warehousing for effective data management.                         |
| CO2      | To classify data warehousing architecture, modeling techniques, ETL processes, and OLAP operations for efficient data analysis.    |
| CO3      | To explain data preprocessing techniques and transformation for improving data quality.                                            |
| CO4      | To provide fundamental knowledge of data mining concepts, techniques, and applications.                                            |
| CO5      | To introduce classification techniques and equip students with the ability to compare and apply different classifiers effectively. |

| Group A |                                               |     |     |       |
|---------|-----------------------------------------------|-----|-----|-------|
| Que. 1  | Answer the following Question                 | CO  | BTL | Marks |
| a.      | What is noisy data?                           | CO3 | 2   | 2     |
| b.      | Define pattern mining.                        | CO4 | 2   | 2     |
| c.      | What is Hadoop?                               | CO5 | 2   | 2     |
| Group B |                                               |     |     |       |
| Que. 2  | Answer the following Question (Solve any Two) | CO  | BTL | Marks |
| a.      | Explain data transformation techniques.       | CO3 | 2   | 4     |

|        |                                               |     |     |       |
|--------|-----------------------------------------------|-----|-----|-------|
| b.     | Discuss data integration.                     | CO3 | 2   | 4     |
| c.     | Explain data quality issues.                  | CO3 | 2   | 4     |
| Que 3. | Answer the following Question (Solve any Two) | CO  | BTL | Marks |
| a.     | Explain classification vs clustering.         | CO4 | 2   | 4     |
| b.     | Describe data mining process.                 | CO4 | 3   | 4     |
| c.     | Explain issues in data mining.                | CO4 | 3   | 4     |
| Que 4. | Answer the following Question (Solve any Two) | CO  | BTL | Marks |
| a.     | Explain BI applications.                      | CO5 | 2   | 4     |
| b.     | Discuss Big Data vs traditional data.         | CO5 | 3   | 4     |
| c.     | Explain core components of Hadoop.            | CO5 | 4   | 4     |

**Bloom's Taxonomy Level (BTL)-** Remember , Understand , Apply, Analyze , Evaluate , Create